Enterprise Multi-Agent Platform & System User Guide

A Comprehensive Guide to Architecture, Features, Business Utility & Operational Manual

Document Type:	System Capability & User Operation Manual	**Author:**	Athena AI Core Engineering
Target Audience:	Enterprise Leadership, IT Admins, & Operations Teams	**Gateway URL:**	http://localhost (API Port 8000)
Security Standard:	Multi-Tenant RBAC + Tamper-Evident HMAC Vault	**Status:**	**Production Operational**

Executive Overview: *Athena AI is a state-of-the-art Multi-Agent Automation Platform built for enterprise environments. It unites LangGraph multi-agent orchestration, RAG-powered document vectorization (ChromaDB), multimodal OCR document intelligence, a visual drag-and-drop workflow engine, enterprise third-party API integrations (Slack, Teams, Jira, Salesforce, SAP), and real-time telemetry into a single, cohesive, secure platform. This manual details the business ROI, architecture, feature mechanics, and step-by-step instructions for operating Athena AI.*

1. How Athena AI Is Useful to Your Organization

Athena AI is designed to solve critical enterprise bottlenecks: fragmented data silos, repetitive manual workflows, slow document extraction, and lack of centralized control over AI deployments. By deploying Athena AI, organizations achieve significant operational impact:

- ■ **Operational Speed & Productivity (10x Acceleration):** Complex multi-step tasks—such as searching cross-department documents, querying SQL databases, reviewing code, and compiling summaries—are executed automatically in seconds by specialized AI agents overseen by an intelligent Supervisor.
- ■ **Unified Enterprise Memory & Context:** Static unstructured company assets (PDF invoices, contracts, technical specifications, policy files) are transformed into a living, queryable knowledge base using hybrid vector and keyword search with exact source citations.
- ■ **No-Code Workflow Automation:** Domain experts can graphically design complex multi-agent pipelines using the drag-and-drop Workflow Builder, delegating background jobs to distributed Celery task queues without requiring software development bandwidth.
- ■ **Seamless Workplace Integration:** Connects directly with enterprise tools like Slack, Microsoft Teams, Outlook, Jira, Salesforce, and SAP, enabling automated message processing, ticket creation, and system sync.

- ■ **Bank-Grade Security & Governance:** Strict multi-tenant workspace isolation, Role-Based Access Control (RBAC), SSO authentication, automated PII redaction, prompt injection defense, and cryptographic HMAC-SHA256 audit logging satisfy enterprise regulatory compliance (SOC2, GDPR).
- ■ **Full Telemetry & Cost Control:** Real-time WebSocket streaming, token consumption tracking, Prometheus scrapers, and Grafana dashboards ensure complete visibility into latency, execution bottlenecks, and AI expenditure.

2. System Architecture & Technical Stack

Component Layer	Technology Stack	Enterprise Role
API Gateway	Nginx / Docker	Reverse proxy, SSL termination, rate limiting & static asset routing
Backend Core	FastAPI (Python 3.11)	High-concurrency REST & WebSocket API layer with dependency injection
Agent Orchestration	LangGraph & LangChain	Supervisor-driven stateful multi-agent DAG execution and checkpointing
AI Memory Vault	ChromaDB / Vector Store	Document embedding, semantic search, hybrid retrieval, & memory ranking
Frontend UI	React 18 + Vite + Tailwind	Responsive Single-Page Application (SPA) with interactive canvas & dashboards
Task Workers	Celery + Redis	Distributed asynchronous job processing for heavy OCR, parsing & workflows
Relational Store	PostgreSQL / SQLite	Persistent storage for user credentials, RBAC roles, audit logs & sessions
Observability	Prometheus + Grafana	Live metrics scraping, memory profiling, and visual operational dashboards

3. Feature-by-Feature Detailed Breakdown

3.1 Multi-Agent Orchestration & AI Supervisor

The core intelligence engine uses a Supervisor pattern built on LangGraph. When a user submits a query, the Supervisor analyzes intent and dynamically routes tasks to specialized agents:

- Research Agent: Performs web searches, data aggregation, and fact verification.
- Document Agent: Parses PDFs, extracts tabular data, and performs deep document comprehension.
- SQL Agent: Translates natural language into sanitized SQL queries, executing read-only database analytics.
- Code Agent: Writes, reviews, and executes Python code in a secure execution sandbox.
- Workflow Agent: Triggers complex multi-step background API pipelines.

3.2 AI Memory Vault & Hybrid RAG Retrieval

Stores and indexes organizational knowledge using state-of-the-art vector embeddings in ChromaDB. Features metadata filtering (by department, tags, user permissions), semantic distance ranking, and hybrid retrieval (combining vector similarity with keyword BM25 scoring). Every response provides transparent source citations with exact page numbers and snippet references.

3.3 Multimodal Document AI & OCR Engine

Processes non-textual assets including scanned PDF invoices, financial receipts, diagrams, and images. Powered by Vision LLMs and Tesseract OCR, the engine performs automatic Named Entity Recognition (NER), structural table extraction, and document classification with high accuracy.

3.4 Visual Drag-and-Drop Workflow Builder

An interactive canvas allowing users to assemble automated workflows without writing code. Users connect Trigger Nodes (e.g., File Upload, Webhook, Schedule), Agent Processing Nodes, Logic Gates (If/Else, Loops), API Connectors, and Action Outputs. Executions are dispatched to distributed Celery background workers.

3.5 Enterprise API Hub & Third-Party Integrations

Native integration connectors for enterprise software tools including Slack, Microsoft Teams, Outlook, Jira, Salesforce, and SAP. Enables agents to read incoming channel messages, summarize email threads, create Jira tickets, update CRM opportunities, or trigger SAP transactions directly.

3.6 Security, RBAC & HMAC Audit Vault

Includes Enterprise SSO (Azure Entra ID / OIDC), granular Role-Based Access Control (Owner, Admin, Manager, Developer, Viewer), automated PII redaction (masking SSNs, credit cards, emails), prompt injection filtering, and a tamper-evident audit logging system secured with HMAC-SHA256 signatures.

3.7 Real-Time Telemetry & System Observability

Streams live execution states via WebSockets to the UI, showing step-by-step agent reasoning and node traversals. Fully instrumented with Prometheus endpoints and pre-configured Grafana dashboards (Port 3001) for real-time memory usage, token throughput, and worker queue depths.

3.8 Document Classifier & Prompt Studio

Provides an automated document categorization panel alongside a Prompt Engineering Studio. Prompts can be tested, version-controlled using MLflow, and benchmarked for accuracy and toxicity before deployment to production.

4. Step-by-Step User Manual & Operating Guide

Step 1: Accessing the Application Portal

1. Open a modern web browser (Chrome, Firefox, Safari, Edge).

2. Navigate to: **http://localhost** (or http://localhost:8000 for raw API docs).

3. You will be greeted by the Athena AI authentication screen.

4. Enter your employee credentials or select 'Sign in with SSO' (Azure Entra ID).

Step 2: Uploading & Managing Documents in the Memory Vault

1. Click on **Memory Vault** or **Document AI** in the left navigation sidebar.

2. Click the **Upload Document** button or drag and drop PDF, PNG, JPG, or TXT files onto the drop zone.

3. Fill in document metadata: Select *Department* (e.g., Legal, Finance, Engineering) and add tags.

4. Click **Process File**. The background Celery worker will perform chunking, vector embedding, and OCR extraction.

5. Once complete, your document status will display **Indexed & Active**.

Step 3: Interacting with the Multi-Agent Chat Interface

1. Navigate to **AI Workspace / Chat** from the navigation sidebar.

2. Type your question or prompt in the bottom message bar (e.g., *'Summarize the Q3 financial report and list top 3 risks'*).

3. Watch the real-time reasoning stream: The **Supervisor Agent** delegates to the **Document Agent** and **Research Agent**.

4. Review the generated response alongside source citations and confidence scores.

5. Use the **Copy** or **Export Report** buttons to save outputs.

Step 4: Building & Executing Visual Workflows

1. Select **Workflow Builder** from the sidebar menu.

2. Drag a **Trigger Node** (e.g., 'On Document Upload') onto the grid canvas.

3. Drag an **Agent Node** (e.g., 'OCR + Entity Extraction Agent') and connect the ports with a directional arrow.

4. Add an **Action Node** (e.g., 'Post Summary to Slack #announcements').

5. Click **Save & Deploy Workflow**. Test the pipeline using the 'Run Test Execution' trigger.

Step 5: Configuring Enterprise Integrations

1. Go to **Settings & Integrations** -> **Integration Panel**.

2. Click **Configure** next to your desired service (Slack, Teams, Jira, Salesforce, SAP).

3. Input the required OAuth tokens or API credentials (saved securely in Secrets Vault).

4. Toggle the integration state to **Active**.

Step 6: Monitoring System Telemetry & Grafana

1. Navigate to **Performance Dashboard** or access Grafana directly at **http://localhost:3001**.

2. Default login credentials: User admin **|** Password admin.

3. View real-time panels for System Memory, CPU Usage, Active Celery Tasks, Token Rate, and Latency.

5. System Troubleshooting & Operations Guide

Symptom / Issue	Probable Cause	Resolution Step
Document stuck in 'Processing'	Celery worker or Redis down	Run docker-compose restart worker redis
504 Gateway Timeout on Chat	LLM API rate limit or model timeout	Check API Key in .env and inspect logs/backend.log
Grafana dashboard empty	Prometheus target unassigned	Verify Prometheus target state at http://localhost:9090/targets
Permission Denied on file upload	Insufficient RBAC permissions	Ask Org Admin to update your user role to *Manager* or *Admin*

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• Technical Documentation: docs/ARCHITECTURE.md
• API Specification: http://localhost/api/v1/docs
• Operations Support: Contact your designated Enterprise System Administrator.